

Project Title: SafeRoute – Personal Safety and Smart Navigation Application

1. Description

SafeRoute is a mobile-based personal safety and smart navigation application designed primarily to enhance women's safety during travel. The application focuses on helping users avoid potentially unsafe areas by analyzing risk levels and suggesting safer routes instead of just the shortest ones.

The system uses real-time location tracking, map integration, and intelligent risk detection to provide users with awareness about their surroundings. In emergency situations, the application enables users to quickly trigger an SOS alert, which sends their live location to trusted contacts via SMS. The goal of this project is to provide both preventive safety (avoiding danger) and reactive safety (quick help during emergencies), making it a practical and impactful real-world solution.

2. Features

- User Registration and Login for secure access
 - Real-Time Location Tracking using GPS
 - Live Map displaying current position
 - Risk Detection (Low, Medium, High zones)
 - Danger Zone Visualization on map
 - Safe Route Navigation (based on safety, not distance)
 - SOS Emergency Alert with location sharing
 - Trusted Contacts Management
 - Smart Alerts when entering risky areas
 - Voice-based SOS trigger for hands-free use
 - Silent SOS for discreet emergency activation
 - Group Feature (up to 4 members with live tracking)
 - Route Risk Scoring and comparison
 - Location-based notifications and warnings
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3. Technology Stack

- Frontend (Mobile Application):
- React Native
- Backend:
- Node.js
- Express.js
- Database:

- MongoDB Atlas
 - APIs and Services:
 - Google Maps API (Maps and Directions)
 - SMS Service (Twilio or Native SMS)
 - Geolocation Services
 - Speech Recognition API
 - Tools and Utilities:
 - JWT for authentication
 - Axios for API communication
 - Development Environment:
 - Visual Studio Code
 - Android Studio (for emulator and testing)
 - Git and GitHub for version control
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4. Workflow of the Project

- User opens the application and logs in or registers
 - The system fetches the user's real-time location
 - The map interface displays current location and nearby risk zones
 - User enters destination location
 - The system analyzes multiple routes and suggests the safest path
 - During travel, the app continuously monitors location and provides alerts for unsafe areas
 - If an emergency occurs, the user can trigger SOS using button, voice, or quick action
 - SOS sends live location to trusted contacts or group members
 - Group members can track the user in real-time and respond quickly
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5. Conclusion

SafeRoute is a practical and impactful solution aimed at improving personal safety, especially for women. Unlike traditional navigation systems, this application focuses on safety-aware routing and proactive risk detection. It not only helps users avoid dangerous areas but also ensures immediate assistance during emergencies through multiple SOS mechanisms.

The project stands out due to its real-world relevance, combination of modern technologies, and balanced approach between usability and innovation. With future enhancements such as advanced AI-based predictions and real-time data integration, this system can be further developed into a full-scale safety platform.